VAMSI KATTERA

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Professional Summary

AI Engineer (MSc Advanced Computer Science, University of Strathclyde) who builds, evaluates and deploys LLM and computer vision applications end to end. Built a GOV.UK RAG assistant with hybrid retrieval, reranking, measured evaluation and a tested, Dockerised FastAPI service deployed on Hugging Face Spaces. Also delivered real-time computer vision and deep learning applications deployed with Docker, Render and AWS. Seeking AI / ML engineer roles building production LLM and RAG applications.

Education

MSc Advanced Computer Science – University of Strathclyde **Jan 2025 – Jan 2026**
Relevant Coursework: Machine Learning, Deep Learning, Artificial Intelligence
BSc Computer Science & Engineering – KHIT, Andhra Pradesh, India **2020 – 2024**
CGPA: 7.66 | Relevant Coursework: Python, Data Structures & Algorithms, AI, ML

Skills

Languages: Python, SQL
LLM & RAG: RAG pipelines, hybrid retrieval (BM25 + dense), cross-encoder reranking, embeddings, prompt engineering, grounded answers with citations, LLM evaluation (LLM-as-judge), Gemini API, OpenAI API, agent workflows and tool integration
Vector Search: Chroma, sentence-transformers
Machine & Deep Learning: TensorFlow, Keras, scikit-learn, CNNs, transfer learning, RNNs, LSTMs, Transformers, model evaluation, hyperparameter tuning, feature engineering
Computer Vision: YOLOv8, OpenCV, object detection, object tracking, video analytics
Backend & APIs: FastAPI, REST API design, request validation, rate limiting, structured logging, metrics
MLOps & Deployment: Docker, GitHub Actions CI, pytest, Hugging Face Spaces, AWS EC2, Render
Tools: Git, GitHub, Streamlit, NumPy, Pandas
Foundations: Linear Algebra, Calculus, Probability & Statistics

AI Projects

GOV.UK RAG Assistant – Cited Q&A over UK Government Guidance | Live Demo | GitHub
Tech Stack: Python, FastAPI, Gemini API, Chroma, BM25, cross-encoder reranking, Streamlit, Docker, GitHub Actions, Hugging Face Spaces

- Built a RAG assistant over 129 GOV.UK pages (visas, right to work, benefits) that answers with citations to the source page and refuses when the pages do not cover the question.
- Combined vector and BM25 search with rank fusion and cross-encoder reranking, raising top-1 retrieval accuracy from 74% to 86% on a hand-reviewed 42-question test set.
- Added a similarity gate and a cite-or-abstain rule: 10/10 out-of-scope questions refused and 95% of in-scope answers judged correct (LLM judge, spot-checked by hand).
- Shipped it as a tested FastAPI service (pytest, CI, rate limiting, JSON logs, metrics) in Docker with a Streamlit UI, deployed on Hugging Face Spaces.

- Fixed answers lost at chunk boundaries by adding neighbouring chunks, and numbered citations per page so every [n] in an answer maps to a source the user can open.
- Built an evaluation harness (retrieval, gate-threshold and LLM-judged answer tables) and documented the known failures in the README.

AI Traffic Monitoring System – Real-Time Computer Vision
Tech Stack: Python, YOLOv8, ByteTrack, OpenCV, scikit-learn, FastAPI, React, Docker, Render

- Built a system that detects, tracks and counts vehicles from video or webcam feeds with YOLOv8 and ByteTrack, with lane-zone counts and a traffic density estimate.
- Trained an Isolation Forest to flag unusual traffic patterns and a Random Forest to classify traffic density.
- Counted vehicles crossing a line and per lane zone, and showed live counts, density and anomaly flags on a React dashboard.
- Served the video stream and analytics through a FastAPI service with a React dashboard, containerised with Docker and deployed on Render.

Jackey – Voice Assistant with Wake-Word Detection
Tech Stack: Python, Picovoice Porcupine, Gemini 2.0 Flash API

- Built a voice assistant that listens offline for a wake word using Picovoice Porcupine, then sends the spoken request to Gemini 2.0 Flash and speaks the reply.
- Added memory of tasks and conversation, and actions to open applications and report the time and weather.

Adaptive Coding Practice App – Gemini-Powered Hints
Tech Stack: Python, Streamlit, Gemini API, pandas, AWS EC2

- Built a Streamlit app with 50 coding challenges and 50 quiz questions that runs submitted code against test cases and scores answers.
- Added adaptive difficulty that moves learners up a level after fast, correct answers, and Gemini-generated hints that guide without giving the full solution.
- Deployed as a Streamlit web application on AWS EC2.

AI vs Real Image Classifier – Deep Learning Computer Vision
Tech Stack: Python, TensorFlow, Keras, MobileNetV2, Streamlit, AWS EC2

- Built a CNN classifier using transfer learning (MobileNetV2) to distinguish AI-generated images from real ones, covering preprocessing, training, validation and evaluation.
- Improved generalisation with data augmentation and dropout regularisation, reaching over 80% classification accuracy.
- Deployed as a real-time Streamlit application on AWS EC2 for live inference.

Certifications

Deep Learning Specialization – DeepLearning.AI (Coursera)
Mathematics for Machine Learning and Data Science – DeepLearning.AI (Coursera)